

Ex-1

```
1  #include <iostream>
2  using namespace std;
3
4  template <typename T>
5  T getMin(T a, T b)
6  {
7      return a < b ? a : b;
8  }
9
10 int main()
11 {
12     int a = 10, b = 20;
13     cout << "The smaller number is: " << getMin(a, b) << endl;
14
15     double x = 5.5, y = 3.3;
16     cout << "The smaller number is: " << getMin(x, y) << endl;
17
18     char c1 = 'A', c2 = 'B';
19     cout << "The smaller character is: " << getMin(c1, c2) << endl;
20
21     return 0;
22 }
```

```
The smaller number is: 10
The smaller number is: 3.3
The smaller character is: A
```

Ex-2

```
1  #include <iostream>
2  using namespace std;
3  template <class T>
4  class Calculator
5  {
6      T num1, num2;
7
8  public:
9      Calculator(T a, T b)
10     {
11         num1 = a;
12         num2 = b;
13     }
14     T add()
15     {
16         return num1 + num2;
17     }
18     T sub()
19     {
20         return num1 - num2;
21     }
22     T mul()
23     {
24         return num1 * num2;
25     }
26     T div()
27     {
28         if (num2 == 0)
29         {
30             cout << "Error: Division by zero!" << endl;
31             return 0;
32         }
33         return num1 / num2;
34     }
35 };
```

```
36 int main()
37 {
38     Calculator<int> c1(111, 110);
39     cout << "The sum is: " << c1.add() << endl;
40     cout << "The difference is: " << c1.sub() << endl;
41     cout << "The product is: " << c1.mul() << endl;
42     cout << "The Division is: " << c1.div() << endl;
43     Calculator<double> c2(5.5, 3.3);
44     cout << "The sum is: " << c2.add() << endl;
45     cout << "The difference is: " << c2.sub() << endl;
46     cout << "The product is: " << c2.mul() << endl;
47     cout << "The Division is: " << c2.div() << endl;
48 }
```

```
The sum is: 221
The difference is: 1
The product is: 12210
The Division is: 1
The sum is: 8.8
The difference is: 2.2
The product is: 18.15
The Division is: 1.66667
```

Ex-3

```
1  #include <iostream>
2  using namespace std;
3
4  int main()
5  {
6      int age;
7      cout << "Enter your age: ";
8      cin >> age;
9
10     try
11     {
12         if (age < 18)
13         {
14             throw "You are Not Eligible";
15         }
16         cout << "You are Eligible" << endl;
17     }
18     catch (const char *message)
19     {
20         cout << message << endl;
21         return 0;
22     }
23     cout << "Program is still running..." << endl;
24 }
```

```
Enter your age: 25
You are Eligible
Program is still running...
```

TASK-1

```
1  #include <iostream>
2  using namespace std;
3  template <typename T>
4  T add(T a, T b)
5  {
6      return a + b;
7  }
8  template <typename T>
9  T diff(T a, T b)
10 {
11     return a - b;
12 }
13 template <typename T>
14 T Product(T a, T b)
15 {
16     return a * b;
17 }
18
19 int main()
20 {
21     int a = 10, b = 20;
22     cout << "The sum is: " << add(a, b) << endl;
23     cout << "The difference is: " << diff(a, b) << endl;
24     cout << "The product is: " << Product(a, b) << endl;
25
26     double x = 5.5, y = 3.3;
27     cout << "The sum is: " << add(x, y) << endl;
28     cout << "The difference is: " << diff(x, y) << endl;
29     cout << "The product is: " << Product(x, y) << endl;
30
31     return 0;
32 }
```

```
The sum is: 30
The difference is: -10
The product is: 200
The sum is: 8.8
The difference is: 2.2
The product is: 18.15
```

TASK-2

```
1  #include <iostream>
2  using namespace std;
3  template <class T>
4  class Demo
5  {
6      T *arr;
7  public:
8      void getData(int n)
9      {
10         arr = new T[n];
11         for (int i = 0; i < n; i++)
12         {
13             cout << "Enter Elements " << i + 1 << ": ";
14             cin >> arr[i];
15         }
16     }
17     void display(int n)
18     {
19         cout << "The elements are: ";
20         for (int i = 0; i < n; i++)
21         {
22             cout << arr[i] << " ";
23         }
24         cout << endl;
25     }
26     T max(int n)
27     {
28         T max = arr[0];
29         for (int i = 0; i < n; i++)
30         {
31             if (arr[i] > max)
32             {
33                 max = arr[i];
34             }
35         }
36         return max;
37     }
38 };
```

```
39 int main()
40 {
41     Demo<int> d1;
42     d1.getData(3);
43     d1.display(3);
44     cout << "The maximum element is: " << d1.max(3) << endl;
45 }
```

```
Enter Elements 1: 25
Enter Elements 2: 5
Enter Elements 3: 6
The elements are: 25 5 6
The maximum element is: 25
```

TASK-3

```
1  #include <iostream>
2  using namespace std;
3
4  int main()
5  {
6      int marks;
7      cout << "Enter your marks: ";
8      cin >> marks;
9      try
10     {
11         if (marks < 0 || marks > 100)
12         {
13             throw "Invalid Marks! Marks Should be between 0-100.";
14         }
15         cout << "Your marks are: " << marks << endl;
16     }
17     catch (const char *message)
18     {
19         cout << message << endl;
20         return 0;
21     }
22     cout << "Program is still running..." << endl;
23 }
```

```
Enter your marks: 45
Your marks are: 45
Program is still running...
-----
```

COMBINED TASK

```
1  #include <iostream>
2  using namespace std;
3  template <class T>
4
5  class Calculator
6  {
7      T num1, num2;
8
9  public:
10     Calculator(T a, T b)
11     {
12         num1 = a;
13         num2 = b;
14     }
15     T add()
16     {
17         return num1 + num2;
18     }
19     T sub()
20     {
21         return num1 - num2;
22     }
23     T mul()
24     {
25         return num1 * num2;
26     }
27
28     void div()
29     {
30         try
31         {
32             if (num2 == 0)
33             {
34                 throw "Cannot divide by zero!";
35             }
36             cout << "The Division is: " << num1 / num2 << endl;
37         }
38         catch (const char *message)
39         {
40             cout << message << endl;
41         }
42     };
43     int main()
44     {
45         Calculator<int> c1(111, 0);
46         cout << "The sum is: " << c1.add() << endl;
47         cout << "The difference is: " << c1.sub() << endl;
48         cout << "The product is: " << c1.mul() << endl;
49         c1.div();
50         Calculator<double> c2(5.5, 3.3);
51         cout << "The sum is: " << c2.add() << endl;
52         cout << "The difference is: " << c2.sub() << endl;
53         cout << "The product is: " << c2.mul() << endl;
54         c2.div();
55     }
```

```
The sum is: 111
The difference is: 111
The product is: 0
Cannot divide by zero!
The sum is: 8.8
The difference is: 2.2
The product is: 18.15
The Division is: 1.66667
-----
```